Seminar 1

Preferences and utility · Problem sheet

Give a short justification for each answer. For a false claim, specify a counterexample. Unless stated otherwise, quantities are nonnegative and topology is relative to the stated domain. No Lagrangian methods are needed.

1. Separability

Opening questions

Write a bundle as (y, z), where $y = (t, c)$ denotes tea and coffee and z is another good.

(a) What does it mean that preferences over y are separable from z? State the condition using comparisons of (y, z) and (y', z) at different fixed values of z.

(b) For $u(t, c, z) = \sqrt{tc} + \log(1 + z)$, find an index $v(t, c)$ and an aggregator $U(v, z)$. Does separability require utility to be additively separable?

(c) For $w(t, c, z) = t + zc$, compare $y = (2, 0)$ and $y' = (0, 2)$ at $z = 1/2$ and at $z = 2$. Is the food group separable from z?

(d) Does a stable ranking within the food group imply a fixed food budget?

2. Quasilinearity

Opening questions

Let $a > 0$, $b > 0$. Consider $u = a + \sqrt{b}$, $v = e^{a+\sqrt{b}}$ and $w = a^2 + b$.

(a) Define a utility function quasilinear in a. Which of these *given functions* have that form?

(b) Which rankings in this list admit a representation quasilinear in a? For w, compare $(a, b) = (1, 4)$ and $(2, 1)$, then add 1 to the a coordinate of both bundles.

(c) For u, how does an indifference curve change when its utility label rises by 1? Is "no income effect on b" valid at every income if $a \geq 0$ is imposed?

3. A tolerance rule

On $\mathbb{R}^2_+$, let $x \succeq y$ iff $x_1 + x_2 \geq y_1 + y_2 - 1$. Is the relation complete? Is it transitive? Can a utility function represent it?

4. AND versus OR

Compare $x \succeq_A y \iff (x_1 \geq y_1 \text{ and } x_2 \geq y_2)$ with $x \succeq_B y \iff (x_1 \geq y_1 \text{ or } x_2 \geq y_2)$. Which are complete? Which are transitive?

5. Indifference from a primitive relation

Define $\sim$ and $\succ$ using only $\succeq$. If two bundles are incomparable, are they indifferent? For rational preferences, prove that $x \sim y$ and $y \sim z$ imply $x \sim z$.

6. Rational people, cyclic majority

Three voters rank $a \succ b \succ c$, $c \succ a \succ b$ and $b \succ c \succ a$. Compute the pairwise majority ranking. Is there a contradiction with individual rationality?

7. A broken limit

Lexicographic preferences on $[0,1]^2$ first compare good 1, then good 2. Are they continuous? Give a sequence of weak comparisons whose limiting comparison fails.

8. Draw the right set

For $u(x) = \min\{2x_1, x_2\}$, draw the indifference set through $(1,2)$ and its weak upper contour set. Are preferences convex? Strictly convex?

9. Change the labels

For $u = x_1 + x_2$, which of e^u, $-e^{-u}$ and $\min\{u,1\}$ represent the same preferences? Can the same preferences have a convex and a nonconvex utility representation?

10. A good that does not matter

On $\mathbb{R}^2_+$, let $u = x_1$. Are preferences continuous, monotone, strongly monotone and locally nonsatiated? Here monotonicity means that increasing *every* good strictly improves the bundle.

11. A consumer who has enough

For $u = -[(x_1-2)^2 + (x_2-1)^2]$, are preferences convex? Strictly convex? Locally nonsatiated? Name the bundle that settles the last question.

12. A purchase is not a full ranking

A consumer selects a from $\{a,b\}$ and b from $\{a,b,c\}$. Can these be selections from one fixed rational ranking? Would your answer change if the records were the *full* sets of best alternatives, respectively $\{a\}$ and $\{b\}$?

13. One larger problem: food and everything else

A consumer has preferences represented by

$$u(a,t,c) = a + 4(tc)^{1/4}, \qquad (a,t,c) \in \mathbb{R}^3_+.$$

All three prices are 1 and the budget is $a + t + c \le m$, where $m > 0$.

(a) Write $u = U(v(t,c), a)$ using $v(t,c) = \sqrt{tc}$. Is the food group separable? Is utility quasilinear in a?

(b) Draw a food indifference curve for $v = 1$. For a fixed food index $v \ge 0$, show that the cheapest way to obtain it has $t = c = v$ and costs $2v$. You may use $t + c \ge 2\sqrt{tc}$; explain when equality holds.

(c) Explain why any optimum exhausts the budget and buys the food index in a cheapest way. Reduce the problem to choosing $v \in [0, m/2]$.

(d) Find the optimal (a,t,c) for $m = 1$, $m = 2$ and $m = 6$, then for general $m > 0$. You may complete a square after setting $s = \sqrt{v}$.

(e) Does separability fix total food spending? Does quasilinearity eliminate income effects at low incomes? Explain both answers using the solution.